

DUNCO HOSPITAL MANAGEMENT SYSTEM

Complete System Documentation

Version 2.4 | September 2026

Dunco Web Solutions

<https://hmse.duncoweb solutions.co.ke/>

Table of Contents

1. Executive Summary
2. Product Overview
3. System Modules (28)
4. User Roles (21)
5. Sidebar Links (185 tested)
6. Insurance/SHA
7. Security
8. Architecture
9. Installation
10. Configuration
11. Current Status
12. Limitations
13. Demo Access
14. Contact

1. Executive Summary

Dunco HMS is a comprehensive hospital management system built on Laravel 12 / PHP 8.2+ with 133 database tables, 110+ models, 584+ routes, 21 user roles, 137 permissions, and 185 sidebar links.

All modules are fully implemented with real functional pages. Every sidebar link has been tested end-to-end across all 10 user roles with 0 failures (1850/1850 tests passed).

SHA integration code is production-ready but requires EHA credentials from the Digital Health Authority to go live.

2. Product Overview

Component	Technology
Backend	Laravel 12 / PHP 8.2+
Frontend	Blade + Alpine.js + Tailwind CSS
Database	MySQL 8.0+ (133 tables)
Auth	Laravel Sanctum + Spatie Permission
PDF	barryvdh/laravel-dompdf
SMS	Twilio SDK / Africa's Talking / Nexmo
Payments	M-Pesa integration

3. System Modules (28)

#	Module	Status
1	Administration	Implemented
2	Patient Management	Implemented
3	Doctor Management	Implemented
4	Appointments	Implemented
5	Reception / Queue	Implemented
6	Outpatient (OPD)	Implemented
7	Inpatient (IPD)	Implemented
8	Nursing	Implemented
9	Pharmacy	Implemented
10	Laboratory	Implemented
11	Radiology	Implemented
12	Billing & Finance	Implemented
13	Insurance / SHA	Code ready, EHA not configured
14	Human Resources	Implemented
15	Blood Bank	Implemented
16	Ambulance & Emergency	Implemented
17	Reports & Analytics	Implemented
18	Settings	Implemented
19	CMS	Implemented
20	Marketing Suite	Implemented
21	Patient Portal	Implemented
22	Telemedicine	Implemented
23	RFID Management	Implemented
24	IoT Bed Monitoring	Implemented
25	Biometric Security	Implemented
26	AI Features	Implemented
27	EHR (HL7/FHIR)	Implemented
28	Finance Module	Implemented

4. User Roles (21)

Role	Permissions	Access Level
Super Admin	137 permissions	Full access
Hospital Admin	97 permissions	Near-full access
Doctor	31 permissions	Clinical focus
Nurse	23 permissions	Patient care
Receptionist	31 permissions	Front desk + billing
Pharmacist	14 permissions	Pharmacy focus
Lab Technician	14 permissions	Lab focus
Accountant	24 permissions	Finance focus
HR Officer	9 permissions	HR focus
Patient	11 permissions	Self-service

5. Sidebar Test Results

All 185 sidebar links tested end-to-end for all 10 user roles. Result: 1850/1850 passed, 0 failures.

User Role	Links Tested	Result
Super Admin	185	185/185 OK
Hospital Admin	185	185/185 OK
Doctor	185	185/185 OK
Nurse	185	185/185 OK
Receptionist	185	185/185 OK
Pharmacist	185	185/185 OK
Lab Technician	185	185/185 OK
Accountant	185	185/185 OK
HR Officer	185	185/185 OK
Patient	185	185/185 OK

6. Insurance / SHA Integration

The ShaService (467 lines) provides production-ready OAuth2 integration with DHA/EHA. Features include: member verification, eligibility checking, pre-authorization, claims submission, and full audit logging.

Current Status: EHA credentials not configured in production. System falls back to local database. Requires facility registration with SHA/DHA to go live.

7. Security

Control	Implementation
Authentication	Laravel Sanctum + Session-based
Password Hashing	Bcrypt
Authorization	Spatie Permission (21 roles, 137 permissions)
CSRF	Laravel CSRF tokens
XSS	Blade auto-escaping
SQL Injection	Eloquent ORM (parameterized)
Session Security	Database-backed, HTTPS cookies
API Rate Limiting	60 req/min per user
Audit Logging	Spatie ActivityLog
HTTPS	Let's Encrypt SSL

8. System Architecture

User -> Web App (Blade/Alpine.js/Tailwind) -> Laravel Application (PHP 8.2) -> MySQL (133 tables)

External Services: SHA/DHA (EHA), M-Pesa, Twilio SMS, Pusher WebSocket, OpenRouter AI, Zoom

9. Installation

Requirements: PHP 8.2+, MySQL 8.0+, Composer 2.x, Node.js 18+

Steps: git clone, composer install, npm install, cp .env.example .env, php artisan key:generate, php artisan migrate, php artisan db:seed, npm run build

10. Configuration

Key env variables: APP_NAME, APP_URL, DB_*, EHA_*, MPESA_*, TWILIO_*, SANCTUM_STATEFUL_DOMAINS

SHA config: config/eha.php (EHA_ENV, EHA_CLIENT_ID, EHA_CLIENT_SECRET, EHA_FACILITY_ID)

11. Current Implementation Status

Production database: 133 tables, all migrations complete

Demo data: 32 users, 331 patients, 5 doctors, 5 nurses, 403 appointments, 93 invoices, 75 medicines, 60 lab tests, 73 beds, 42 insurance policies

All 185 sidebar links functional. Zero 500 errors. Clean error log.

12. Limitations

Item	Priority	Description
SHA EHA Credentials	High	Not configured; local DB fallback active
M-Pesa Production	High	Using sandbox URL
Twilio SMS	Medium	Needs production credentials
Mail Config	Medium	SMTP placeholder values
.env Security	High	File permissions too permissive

13. Demo Access

URL: <https://hmse.duncoweb solutions.co.ke/>

Role	Email	Password
Admin	admin@duncohms.com	admin123
Doctor	dr.mwangi@duncohms.com	doctor123
Nurse	nurse.njeri@duncohms.com	nurse123
Receptionist	reception.kimani@duncohms.com	reception123
Pharmacist	pharmacist.otieno@duncohms.com	pharmacist123
Lab Tech	lab.wekesa@duncohms.com	labtech123
Accountant	accountant.ogutu@duncohms.com	accountant123
HR Officer	hr@duncohms.com	hr123
Patient	patient.kamau@duncohms.com	patient123

14. Contact

Dunco Web Solutions

Email: dunthecan02@gmail.com

Website: <https://duncoweb solutions.co.ke/>

GitHub: <https://github.com/dunte1/dunco-hms>